

Introduction to Large Language Models

Procheta Sen



What is a Language Model?

- ▶ The classic definition of a language model (LM) is a probability distribution over sequences of tokens.
- ▶ Intuitively, a good language model should have linguistic capabilities and world knowledge.
- ▶ Formally a language model defines:

$$p(x_1, x_2, \dots, x_T)$$

Using the chain rule of probability, this can be decomposed as:

$$p(x_1, x_2, \dots, x_T) = \prod_{t=1}^T p(x_t \mid x_1, \dots, x_{t-1})$$

Fundamental Intuition of Large Language Models

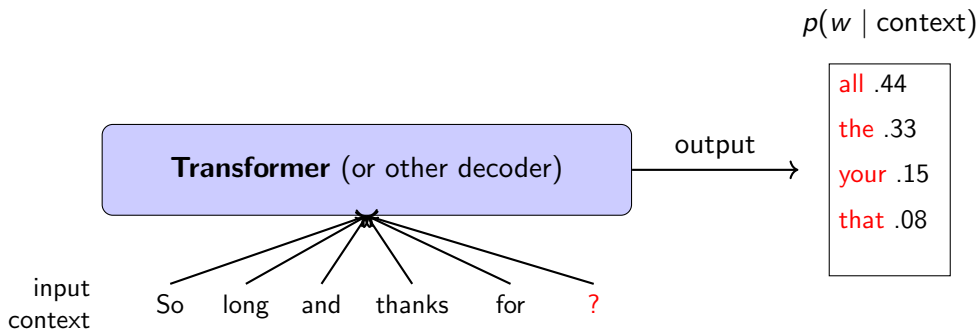
Text contains enormous amounts of knowledge.

Pretraining on lots of text with all that knowledge is what gives language models their ability to do so much.

What is a Large Language Model?

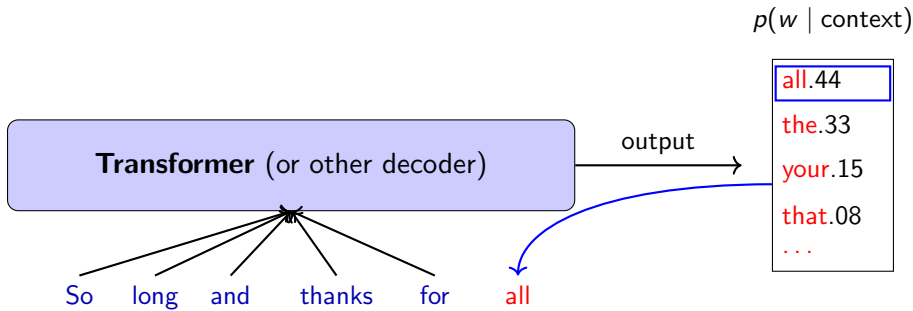
A neural network with:

- ▶ **Input:** a context or prefix
- ▶ **Output:** a distribution over possible next words



LLMs can generate!

A model that gives a probability distribution over next words can generate by repeatedly sampling from the distribution, also known as autoregressive generation.



Big idea

Many tasks can be turned into tasks of predicting words!

Conditional Generation

1. Give the LLM an input piece of text, a **prompt**
2. Have it generate token by token
 - conditioned on the prompt and the generated tokens

We generate from a model by

1. computing the probability of the next token w_i from the prior context: $P(w_i \mid w_{<i})$
2. sampling from that distribution to generate a token

NLP Tasks as Conditional Generation

Many practical NLP tasks can be cast as conditional generation!

Sentiment analysis: “I like Jackie Chan”

1. We give the language model this string:

The sentiment of the sentence ‘‘I like Jackie Chan’’ is:

2. And see what word it thinks comes next

Prompt

Prompt: a text string that a user issues to a language model to get the model to do something useful by conditional generation

Prompt engineering: the process of finding effective prompts for a task.

Prompts

A question:

What is a transformer network?

Perhaps structured:

Q: What is a transformer network? A:

Or an instruction:

Translate the following sentence into Hindi:
'Chop the garlic finely'.

Prompt Structure

Prompts can be very structured

Human: Do you think that “input” has negative or positive sentiment?

Choices:

(P) Positive

(N) Negative

Assistant: I believe the best answer is: (

Prompt Structure

Prompts can have **demonstrations** (= examples)

The following are multiple choice questions about high school computer science.

Let $x = 1$. What is $x \ll 3$ in Python 3?

(A) 1 (B) 3 (C) 8 (D) 16

Answer: C

Which is the largest asymptotically?

(A) $O(1)$ (B) $O(n)$ (C) $O(n^2)$ (D) $O(\log n)$

Answer: C

What is the output of the statement “a + ab” in Python 3?

(A) Error (B) aab (C) ab (D) a ab

Answer:

Prompts are a learning signal

This is especially clear with demonstrations

But this is a different kind of learning than pretraining

- ▶ Pretraining sets language model weights via gradient descent
- ▶ Prompting just changes the context and the activations in the network; no parameters change

We call this **in-context learning**— learning that improves model performance but does not update parameters

LLMs usually have a system prompt

<system> You are a helpful and knowledgeable assistant.
Answer concisely and correctly.

This is automatically and silently concatenated to a user prompt

<system> You are a helpful and knowledgeable assistant.
Answer concisely and correctly. <user> **What is the capital
of France?**

System prompts can be long; 1700 words for Claude Opus4

Some extracts:

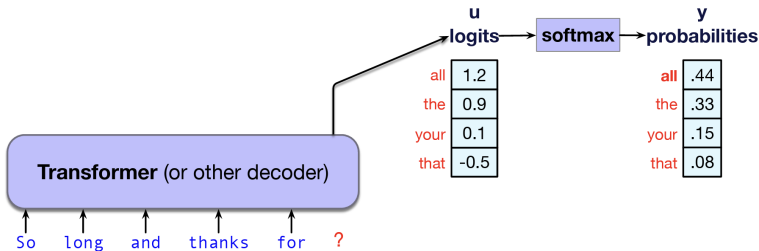
- ▶ Claude should give concise responses to very simple questions, but provide thorough responses to complex and open-ended questions.
- ▶ Claude is able to explain difficult concepts or ideas clearly. It can also illustrate its explanations with examples, thought experiments, or metaphors.
- ▶ Claude does not provide information that could be used to make chemical or biological or nuclear weapons.
- ▶ For more casual, emotional, empathetic, or advice-driven conversations, Claude keeps its tone natural, warm, and empathetic.
- ▶ Claude cares about people's well-being and avoids encouraging or facilitating self-destructive behavior.
- ▶ If Claude provides bullet points in its response, it should use markdown, and each bullet point should be at least 1–2 sentences long unless the human requests otherwise.

Where does token probability come from?

The internal networks for LLMs generate real-valued scores called **logits** for each token in the vocabulary.

A score vector **u** of shape $[1 \times |V|]$ is turned into a probability by softmax:

$$\mathbf{y} = \text{softmax}(\mathbf{u})$$



Decoding

This task of choosing a word to generate based on the model's probabilities is called **decoding**.

Decoding from a model left-to-right and repeatedly choosing the next token conditioned on our previous choices is called **autoregressive generation**.

Greedy Decoding

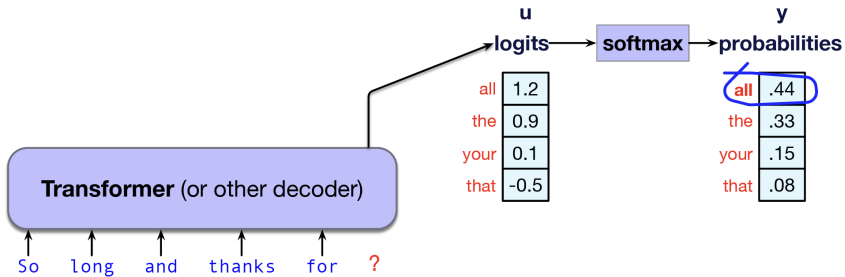
A **greedy algorithm** is one that makes a choice that is locally optimal

- (whether or not it will turn out to have been the best choice with hindsight)

Simply generate the most probable word:

$$\hat{w}_t = \arg \max_{w \in V} P(w \mid \mathbf{w}_{<t})$$

Greedy decoding: choosing “all”



We don't use greedy decoding

Because the tokens it chooses are (by definition) **extremely predictable**, the resulting text is **generic** and **repetitive**.

Greedy decoding is so predictable that it is **deterministic**.

Instead, people prefer text that is more diverse, like that generated by **sampling**.

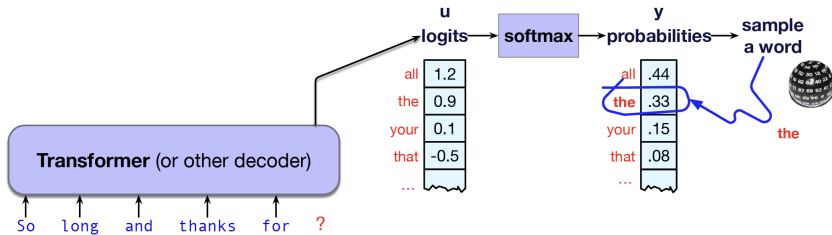
Random sampling

Sampling from a distribution means to choose random points according to their likelihood.

Sampling from an LM means to choose the next token to generate according to its probability.

Random (multinomial) sampling: We randomly select a token to generate according to its probability defined by the LM, conditioned on our previous choices, generate it, and iterate.

Random Sampling: Example



Random sampling

$$i \leftarrow 1$$

$$w_i \sim p(w)$$

while $w_i \neq \text{EOS}$

$$i \leftarrow i + 1$$

$$w_i \sim p(w_i \mid w_{<i})$$

Alas, random sampling doesn't work very well

- ▶ Even though random sampling mostly generates sensible, high-probability words,
- ▶ There are many odd, low-probability words in the **tail** of the distribution
- ▶ Each one is low-probability, but added up they constitute a large portion of the distribution
- ▶ So they get picked often enough to generate **weird sentences**

Factors in word sampling: quality and diversity

Emphasize high-probability words

- ▶ + **quality**: more accurate, coherent, and factual
- ▶ – **diversity**: boring, repetitive

Emphasize middle-probability words

- ▶ + **diversity**: more creative, diverse
- ▶ – **quality**: less factual, incoherent

Temperature sampling

Reshape the probability distribution

- ▶ increase the probability of the high-probability tokens
- ▶ decrease the probability of the low-probability tokens

Temperature Sampling

Divide the logit by a temperature parameter τ before passing it through the softmax.

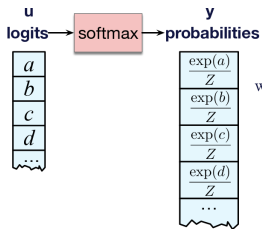
Instead of

$$\mathbf{y} = \text{softmax}(u)$$

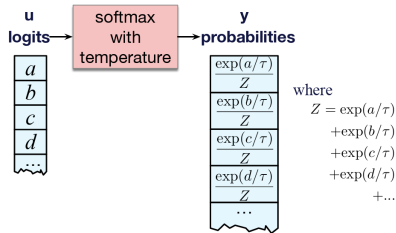
We do

$$\mathbf{y} = \text{softmax}\left(\frac{u}{\tau}\right)$$

Temperature Sampling Example



(a)



(b)

Temperature Sampling

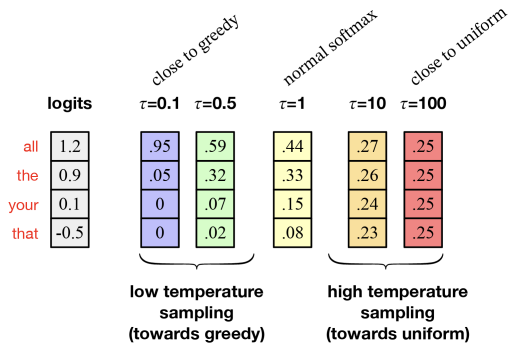
$$0 \leq \tau \leq 1$$

$$\mathbf{y} = \text{softmax}\left(\frac{\mathbf{u}}{\tau}\right)$$

Why does this work?

- ▶ When τ is close to 1, the distribution doesn't change much.
- ▶ The lower τ is, the larger the scores being passed to the softmax.
- ▶ Softmax pushes high values toward 1 and low values toward 0.
- ▶ Large inputs push high-probability words higher and low-probability words lower, making the distribution more greedy.
- ▶ As $\tau \rightarrow 0$, the probability of the most likely word approaches 1.

Softmax Output with Temperature τ



Temperature sampling comes from thermodynamics

- ▶ a system at high temperature is flexible and can explore many possible states,
- ▶ a system at lower temperature is likely to explore a subset of lower energy (better) states.

In low-temperature sampling, ($\tau \leq 1$) we smoothly

- ▶ increase the probability of the most probable words
- ▶ decrease the probability of the rare words.

Introduction to Transformers

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April 19, 2026

What is a Transformer?

- ▶ A deep learning architecture based on *Attention*, introduced in the paper titled 'Attention is All You Need' in NeurIPS 2017.
- ▶ Core of today's NLP models: BERT, GPT, T5, translation.
- ▶ **Background:**

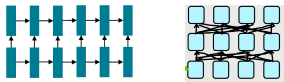


Figure: Sequence to Sequence Processing RNN/LSTM Vs. Transformers

- ▶ RNNs struggled with long Sequence.
- ▶ LSTMs couldn't process tokens in Parallel
- ▶ **Advantages:**
 - ▶ Captures long-range dependencies effectively and trains efficiently in parallel
 - ▶ Achieves state-of-the-art results across NLP tasks.

How Transformer works?

1. Transformer has primarily two components: Encoder, Decoder. Encoder has self attention and decoder has masked self attention.
2. The encoder builds rich, contextual representations of the input sequence
3. *Decoder* The output of decoder is a set of probabilities on a vocabulary set, generates tokens step by step.
4. **Variations in Architecture:** Encoder-only (e.g., BERT) $\Rightarrow$ for understanding tasks , Decoder-only (e.g., GPT) $\Rightarrow$ for text generation, Encoder–Decoder (e.g., T5, original Transformer) $\Rightarrow$ for seq2seq tasks.

Transformer Block Diagram

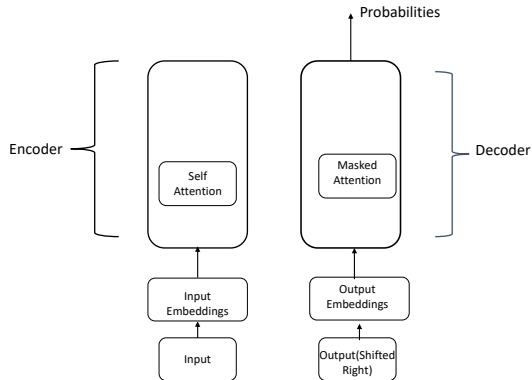


Figure: Transformer Block Diagram

Transformer Architecture Breakdown

- ▶ Both encoder and decoder comprises 6 identical layers as shown in Figure.
- ▶ Three important components: Residual Connection, Layer normalisation and positional encoding .
- ▶ *Positional Encoding* – adds word order information.
- ▶ Directly passing 'raw' embeddings to the next layer can help preventing forgetting/distortion of information to the next layer, residual connections were proposed. $x_{l+1} = F(x_{l-1}) + x_l$
- ▶ layer norm essentially normalizes the output of feed forward layer
- ▶ *Residual Connections + Layer Norm* – stabilize training.

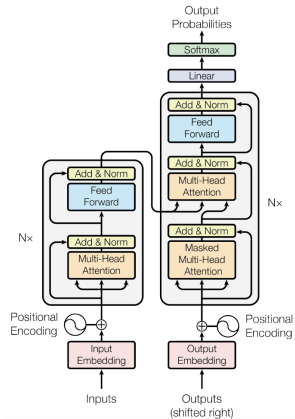


Figure: Vashwani et al. 2017

Deep Dive to Attention Mechanisms: Part 1

- ▶ Attention is Like highlighting important words while reading a paragraph..
Example: In “The cat sat on the mat”, to understand “cat”, the words “sat” and “mat” are more relevant than “the”.
- ▶ **Mechanism (high-level):** Each word builds a query, key, and value. Similarity between query and keys decides which values to “attend to.”
- ▶ **Intuition:** Query: ‘what this position is looking for.’
Key: ‘what each token offers to be matched on.’
Value: Weighted average of the query-key similarity values.

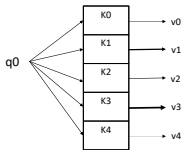


Figure: Overview of Attention

Deep Dive to Attention Mechanisms: Part 2

Scaled Dot-Product Attention Parallelizes the operation. Attention is computed on n number of queries simultaneously

Mathematical Representation:

$$\mathbf{Q} = \mathbf{XW}_Q, \quad \mathbf{K} = \mathbf{XW}_K,$$

$$\mathbf{V} = \mathbf{XW}_V$$

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{QK}^\top}{\sqrt{d_k}}\right)$$

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{AV}$$

- ▶ **Scaling** $1/\sqrt{d_k}$ keeps dot-products numerically stable.
- ▶ **Multi-head**: compute multiple (Q, K, V) in parallel and concatenate.

Example Illustration

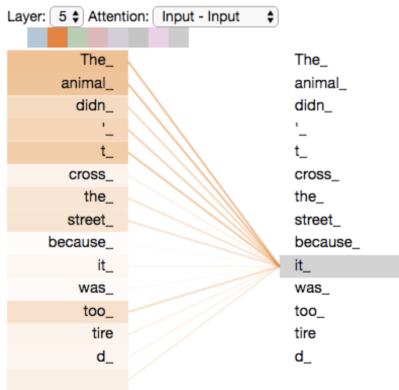


Figure: Attention pattern for the word 'it' in the sentence 'The animal didn't cross the street because it was too tired' (Jay Alammar Transformer Illustrations)

Pretrain / Finetuning

▶ **Pretraining:**

- ▶ Train a neural network to perform language modeling on a large amount of text.
- ▶ Save the network parameters.
- ▶ Captures general linguistic patterns: syntax, semantics, discourse.

▶ **Fine-tuning:**

- ▶ Adapt the pretrained model to a specific downstream task (e.g., sentiment analysis, translation, QA).
- ▶ Requires relatively small labeled datasets.
- ▶ Often only a few epochs are sufficient.

▶ **Benefits:**

- ▶ Dramatic reduction in task-specific data requirements.
- ▶ Faster convergence and higher accuracy.
- ▶ Enables transfer learning across domains and languages.

Important Questions and Bottlenecks

- ▶ **Scalability:** How can we efficiently train ever-larger Transformers (billions of parameters) without prohibitive compute and energy costs?
- ▶ **Data requirements:** Do we always need massive text corpora? How do we adapt models to low-resource languages and domains?
- ▶ **Interpretability:** What exactly do attention weights capture? Can we reliably explain model predictions?
- ▶ **Deployment bottlenecks:** Memory footprint and latency in real-world applications (e.g., mobile devices, chatbots).

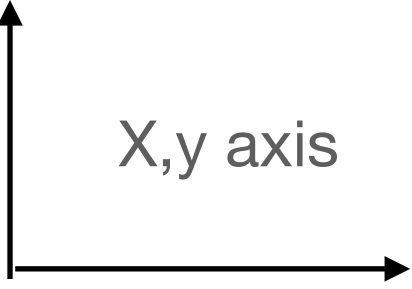
Data Visualization

Procheta Sen

Basic Components of Visualisation

- **Aesthetics**

- *Position:* Describes where the element is located.

- E.g. 

- *Shape:* 

- *Size:* 

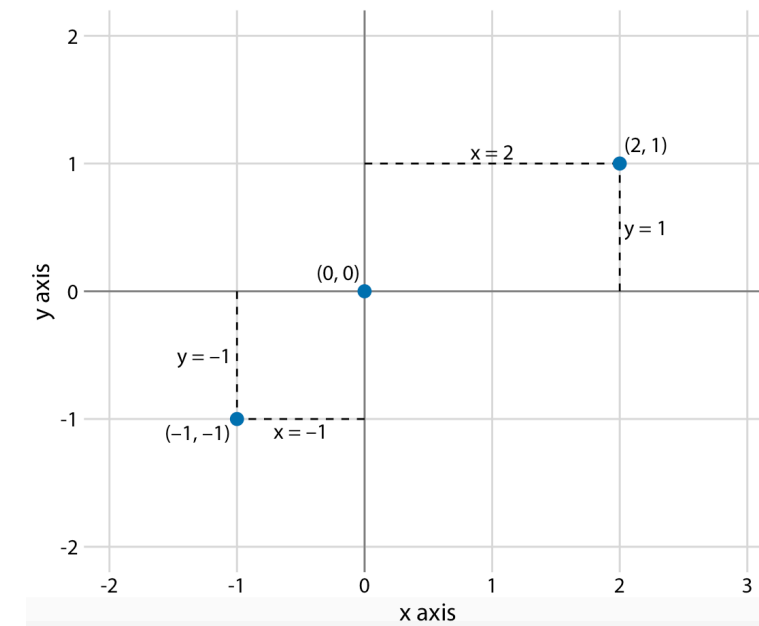
- *Color:* 

- **Types of Data**

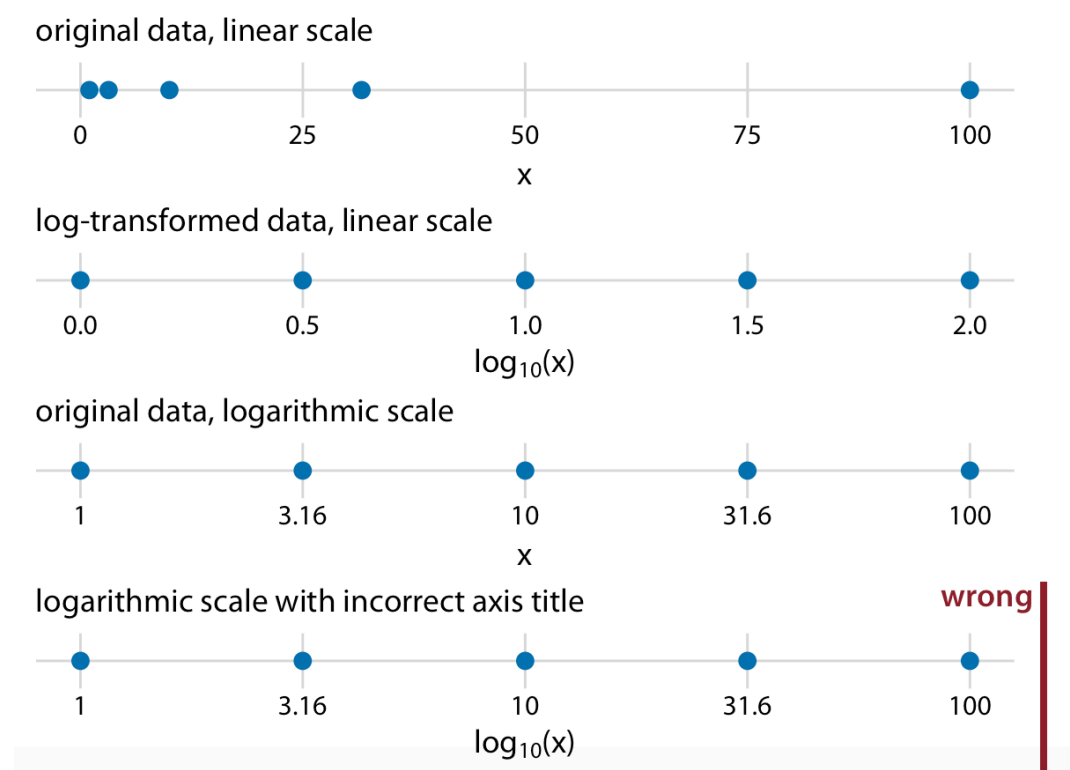
- quantitative/numerical continuous, [1,2,3], [1.5,2.6]
 - numerical discrete, categorical (ordered/unordered), [dog, cat, fish], [good, fair, poor]
 - date/time: Jan. 5 2018, 8:03am
 - Text: The quick brown fox jumps over the lazy dog.

Types of Positions

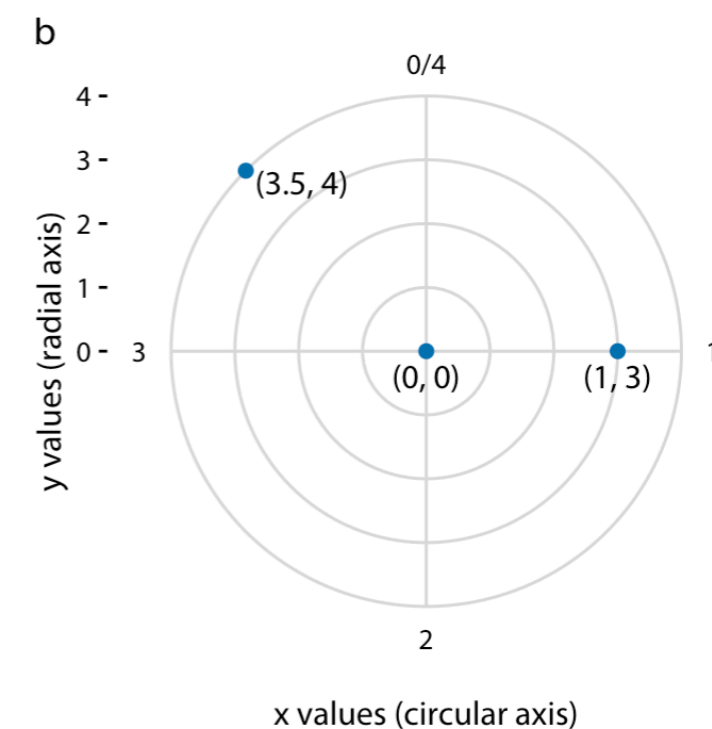
- Cartesian coordinates



- Nonlinear axes



- Coordinate systems with curved axes



Color Scales

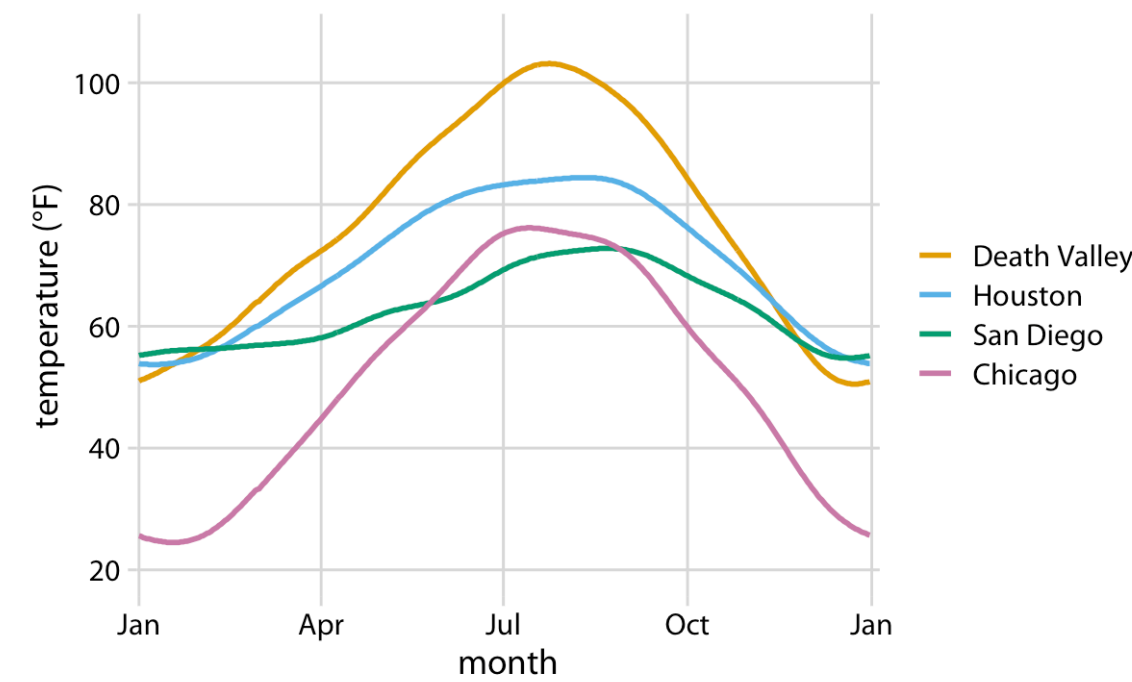
- Many appropriate qualitative color scales are readily available.
- ColorBrewer project provides a nice selection of qualitative color scales, including both fairly light and fairly dark colors (Brewer [2017](#))



Example qualitative color scales. The Okabe Ito scale is the default scale used throughout this book (Okabe and Ito [2008](#)). The ColorBrewer Dark2 scale is provided by the ColorBrewer project (Brewer [2017](#)). The ggplot2 hue scale is the default qualitative scale in the widely used plotting software ggplot2.

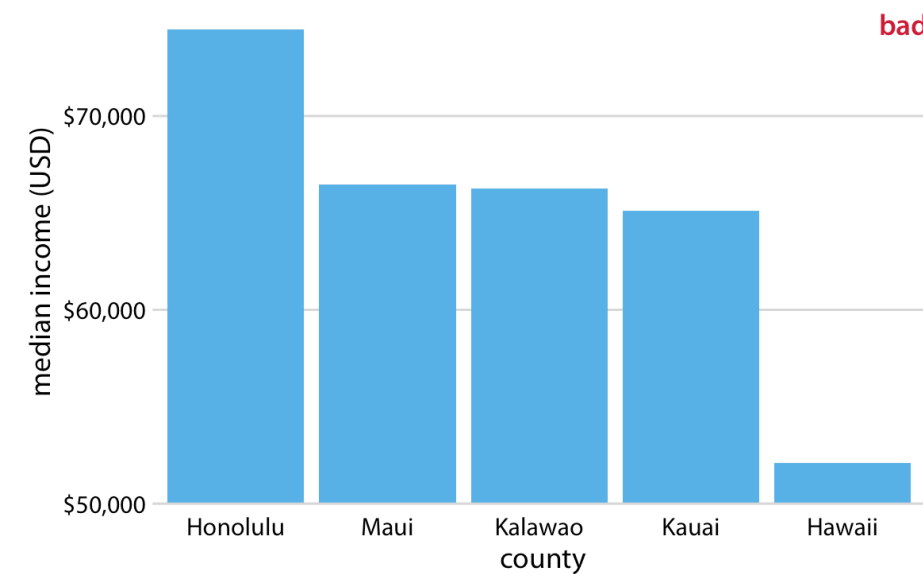
Basic Visualisation Principle

- There is a mapping between data values and aesthetics values. This mapping is done via *scales*.
- A scale must be one-to-one



Daily temperature normals for four selected locations in the U.S. Temperature is mapped to the y axis, day of the year to the x axis, and location to line color. Data source: NOAA.

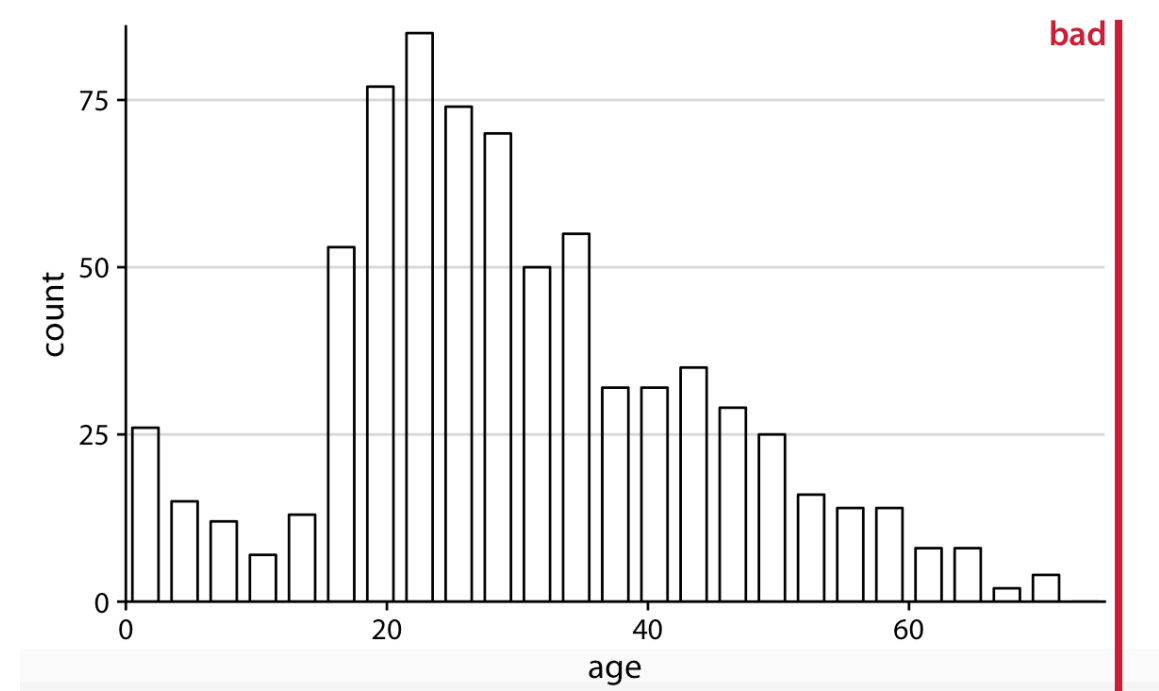
- The sizes of shaded areas in a visualisation need to be proportional to the data values they represent.



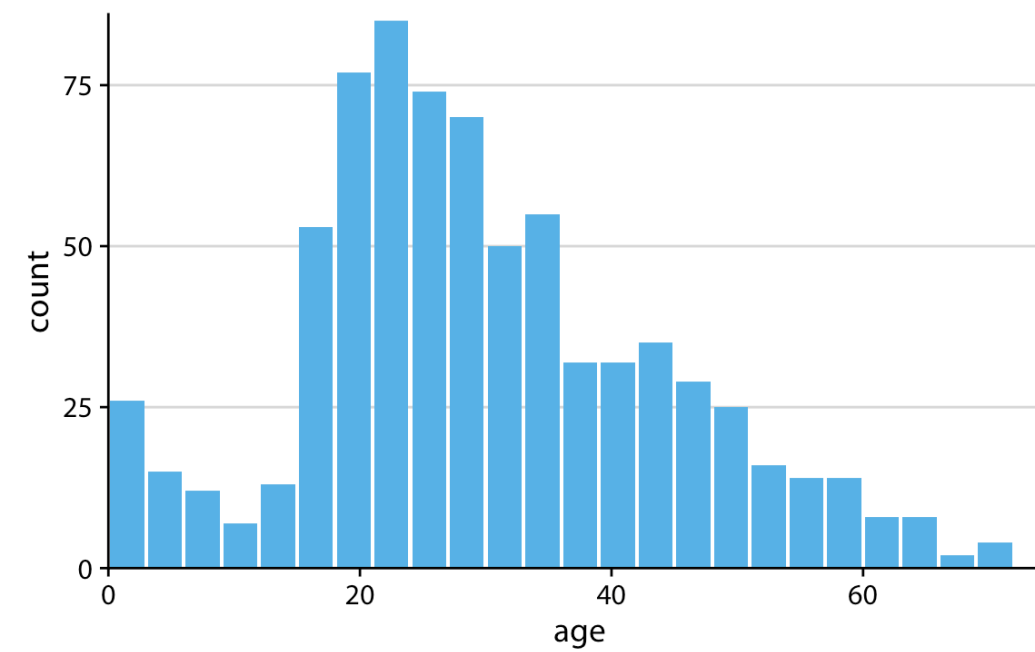
Median income in the five counties of the state of Hawaii. This figure is misleading, because the y axis scale starts at \$50,000 instead of \$0. As a result, the bar heights are not proportional to the values shown and the income differential between the county of Hawaii and the other four counties appears much bigger than it actually is. Data source: 2015 Five-Year American Community Survey.

Basic Visualisation Principle (Continued)

- Use larger axis labels.
- Avoid line drawings.



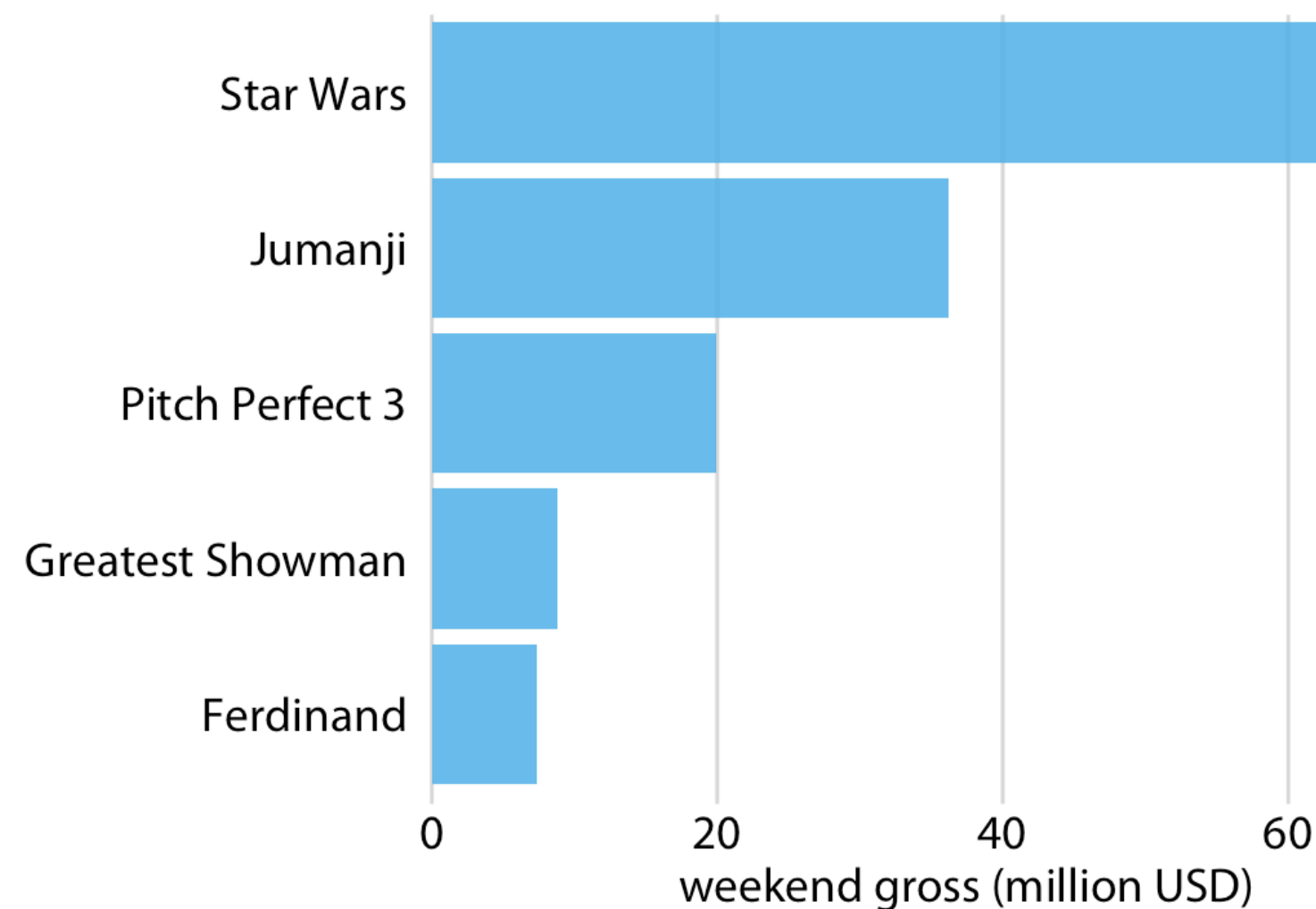
Histogram of the ages of Titanic passengers, drawn with empty bars.



- Choosing the right visualization software.

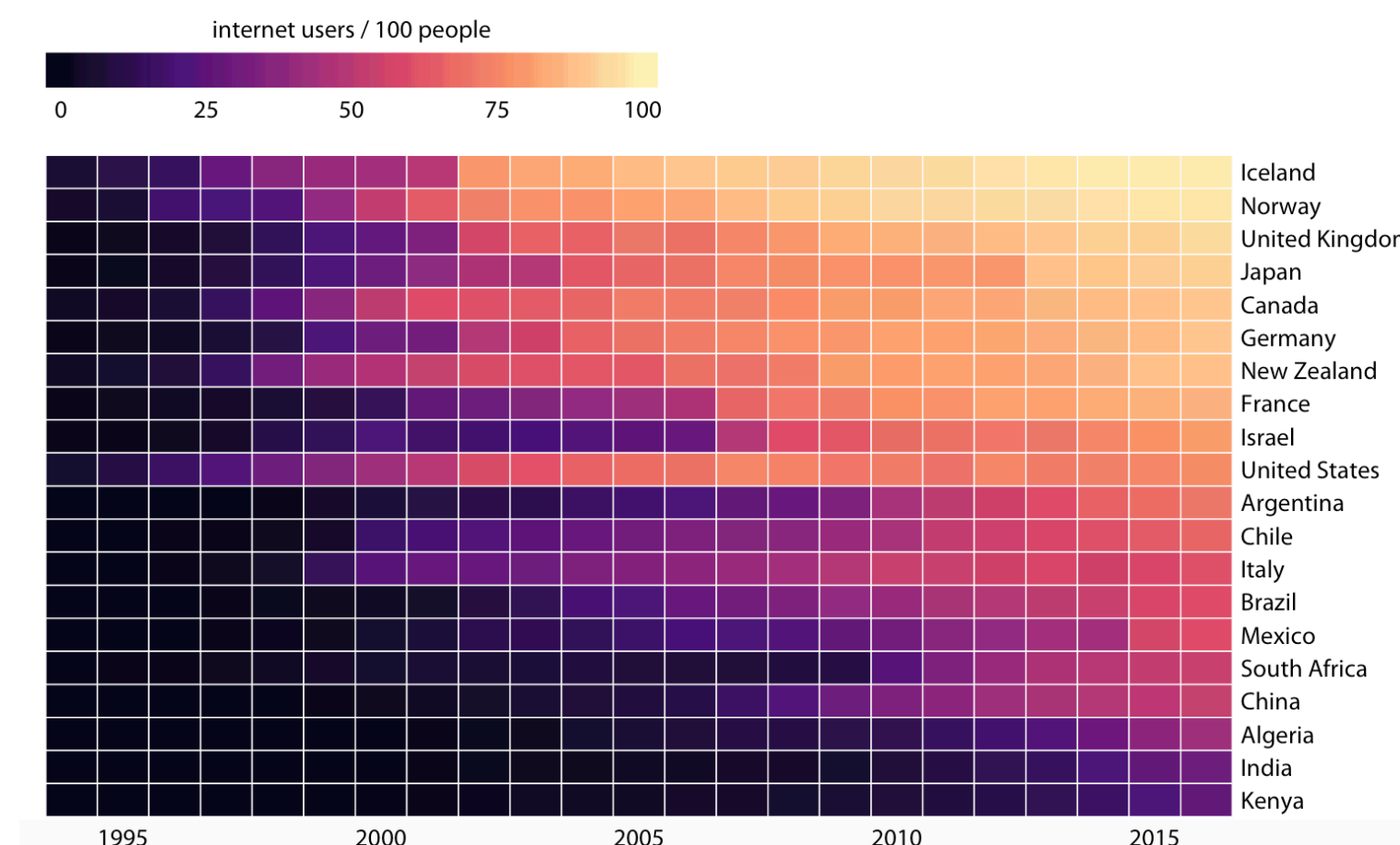
Types of Visualisation: Amount

- Bar Chart



Highest grossing movies for the weekend of December 22-24, 2017, displayed as a horizontal bar plot. Data source: Box Office Mojo (<http://www.boxofficemojo.com/>).

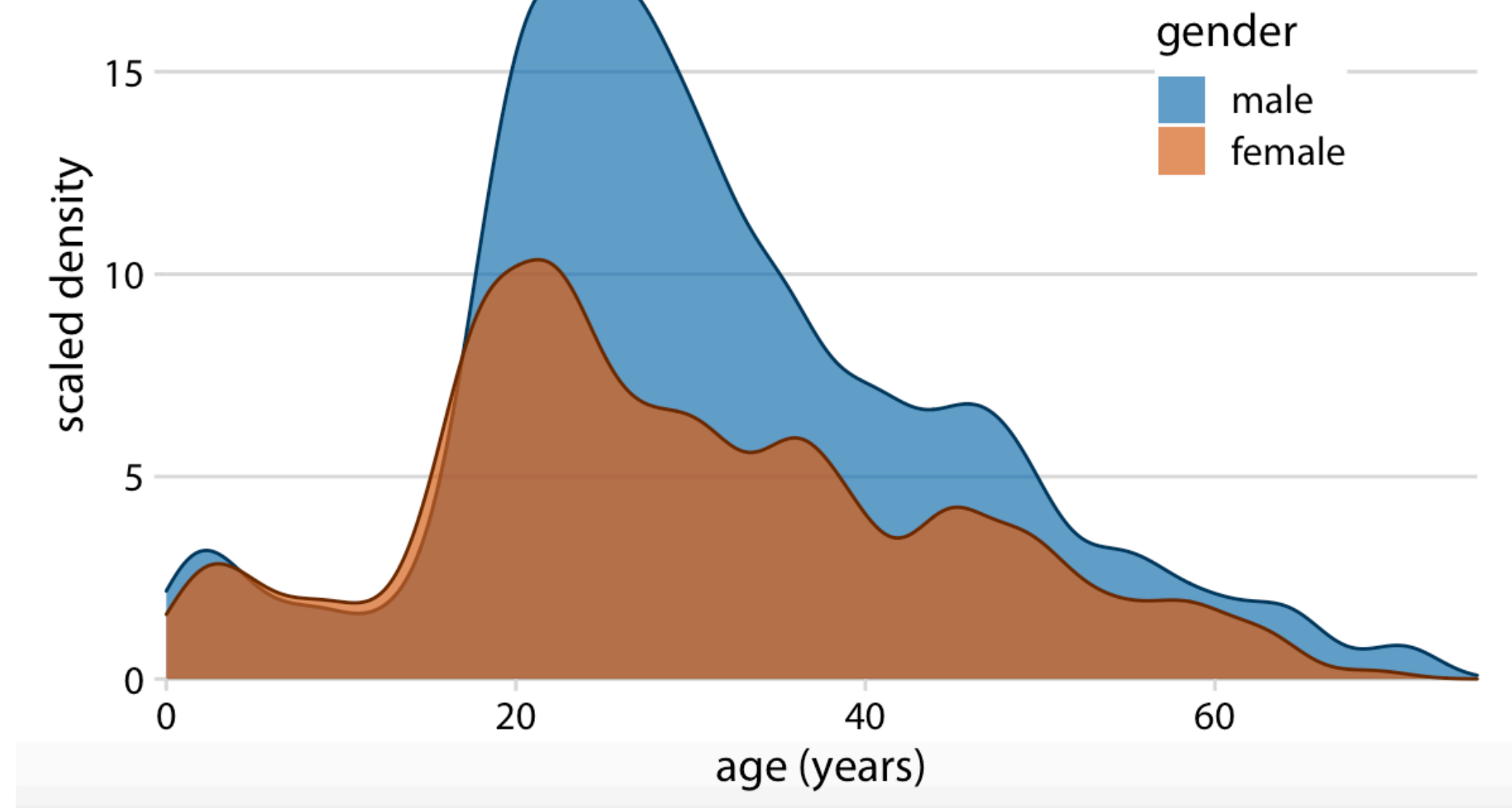
- Heat Map



Internet adoption over time, for select countries. Color represents the percent of internet users for the respective country and year. Countries were ordered by percent internet users in 2016. Data source: World Bank

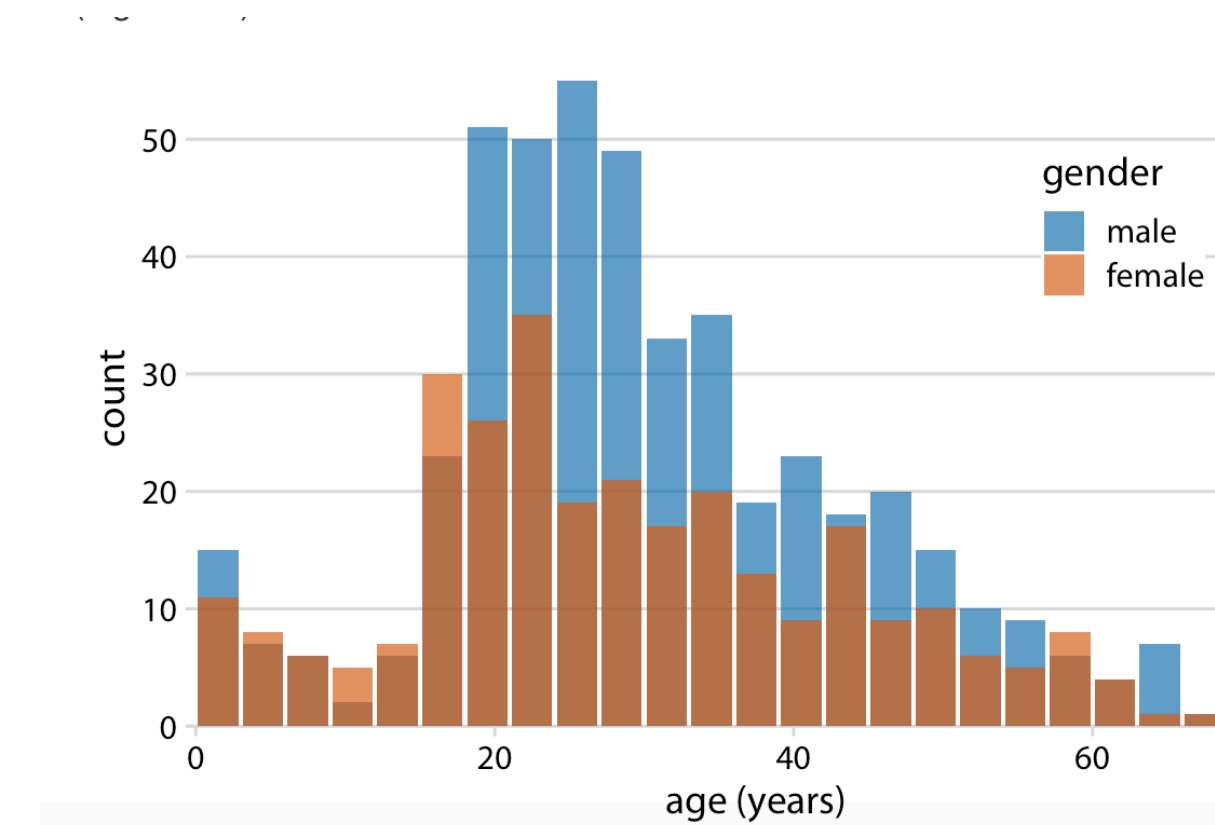
Types of Visualisation: Distribution

- Density Plot



Density estimates of the ages of male and female Titanic passengers. To highlight that there were more male than female passengers, the density curves were scaled such that the area under each curve corresponds to the total number of male and female passengers with known age (468 and 288, respectively).

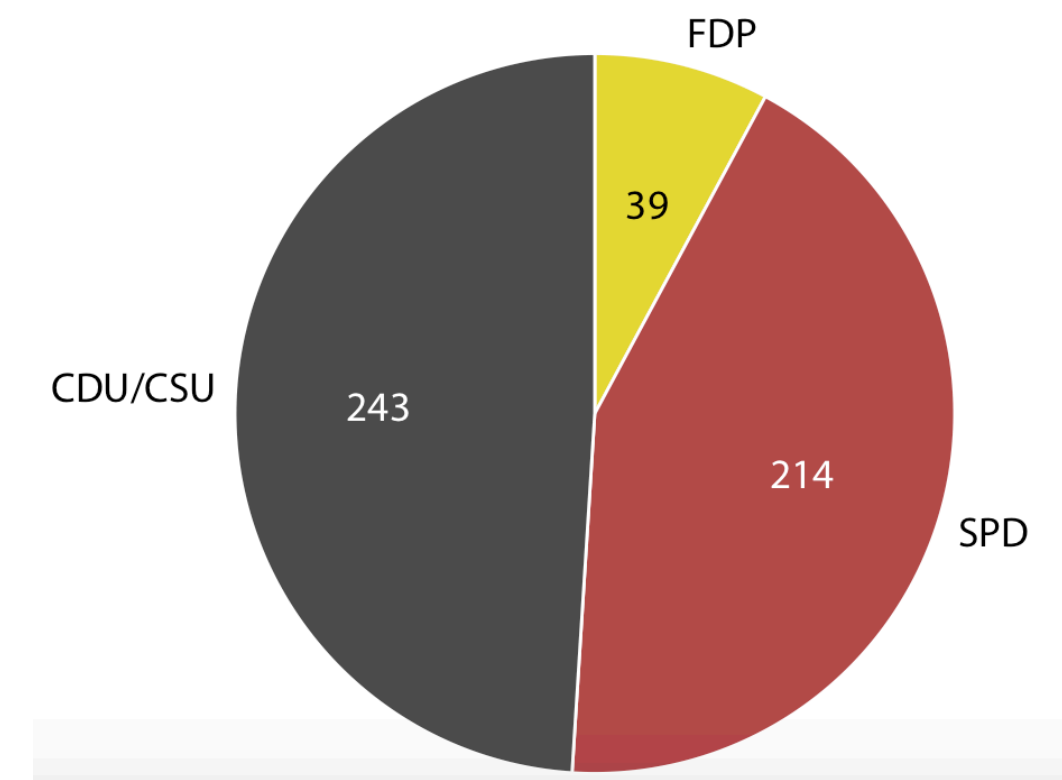
- Histogram



Age distributions of male and female Titanic passengers, shown as two overlapping histograms.

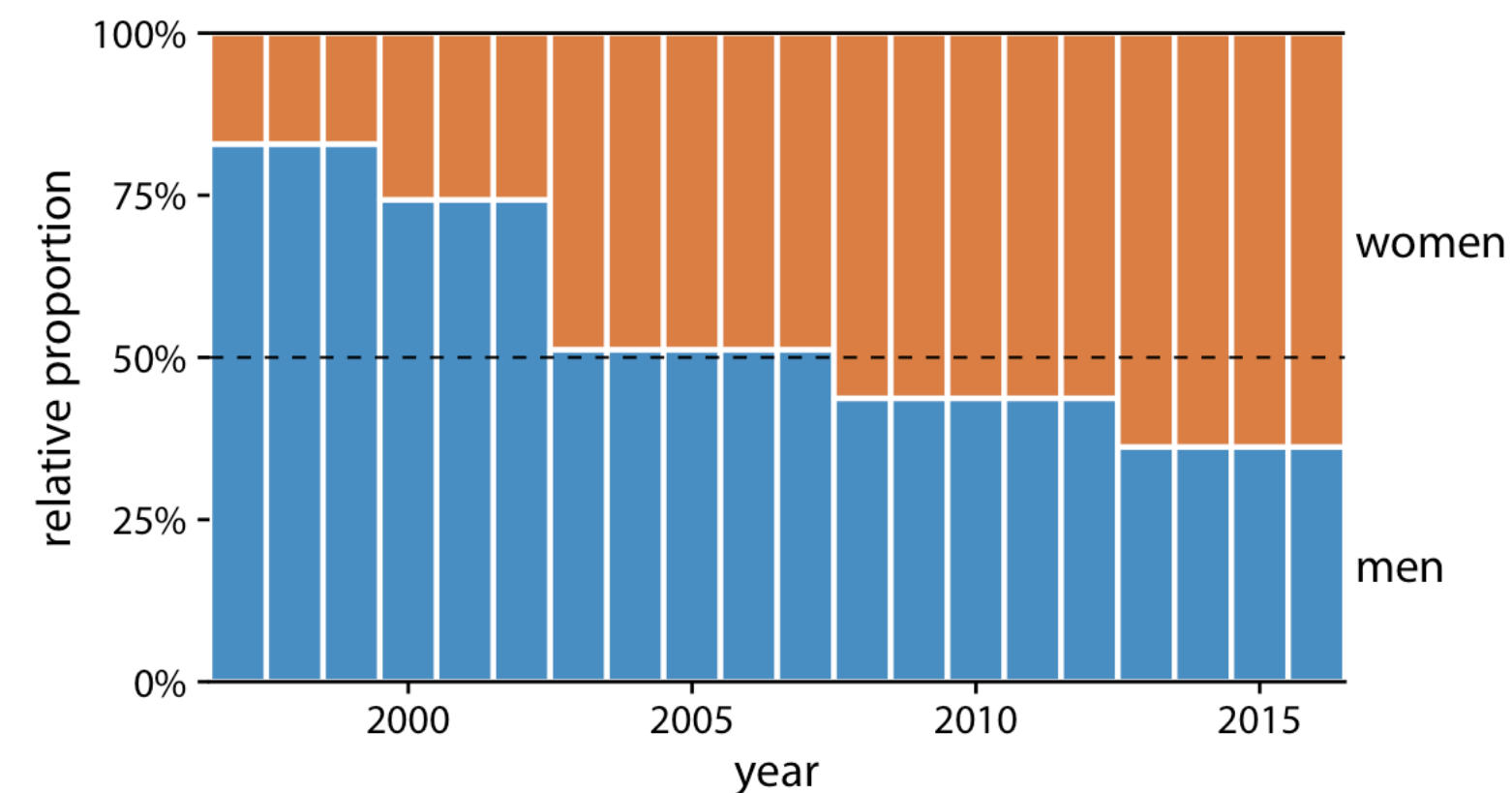
Types of Visualisation: Proportions

- Pie Chart



Party composition of the 8th German Bundestag, 1976–1980, visualized as a pie chart. This visualization shows clearly that the ruling coalition of SPD and FDP had a small majority over the opposition CDU/CSU.

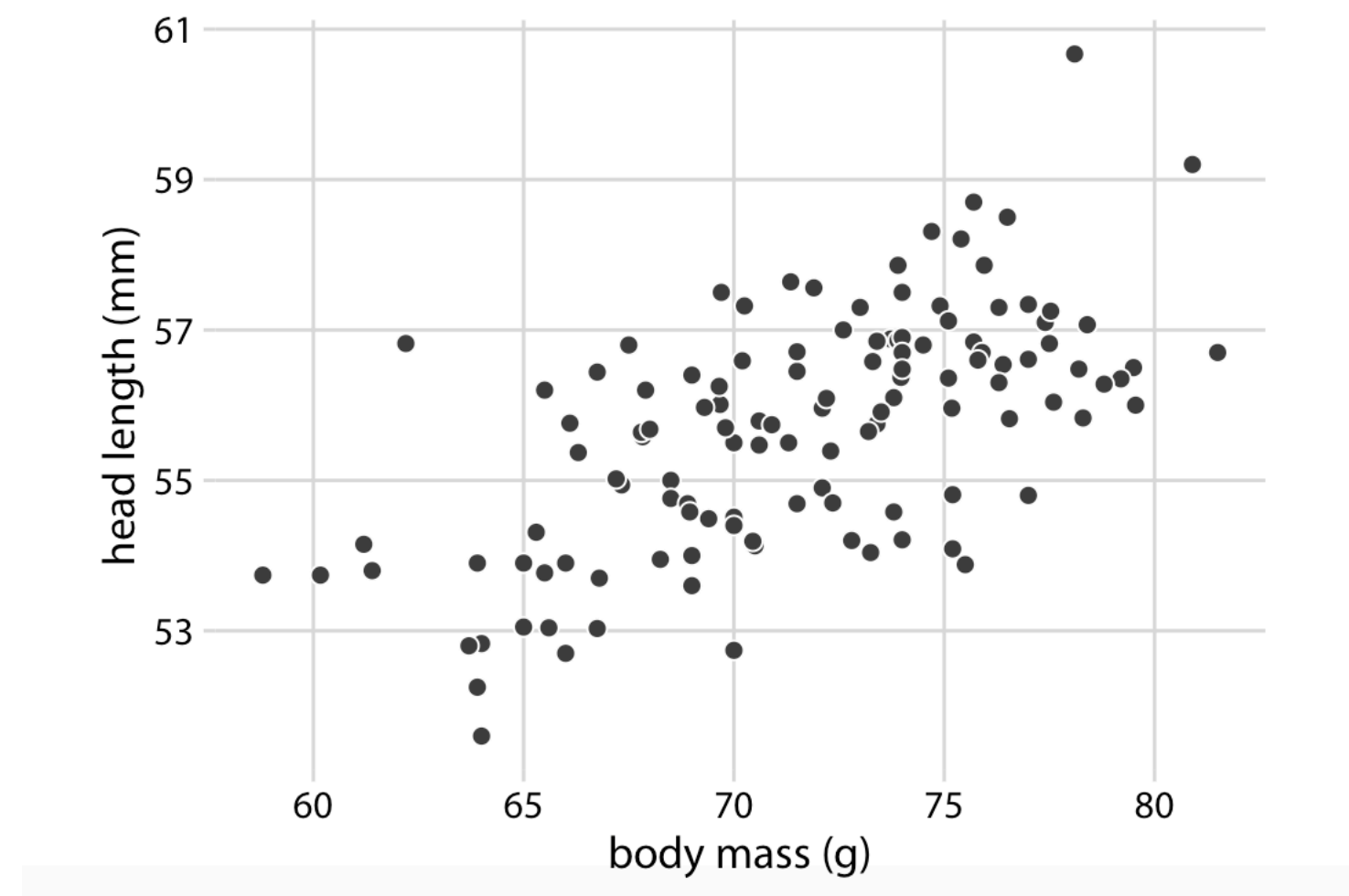
- Side by Side Bars



Change in the gender composition of the Rwandan parliament over time, 1997 to 2016. Data source: Inter-Parliamentary Union (IPU), ipu.org.

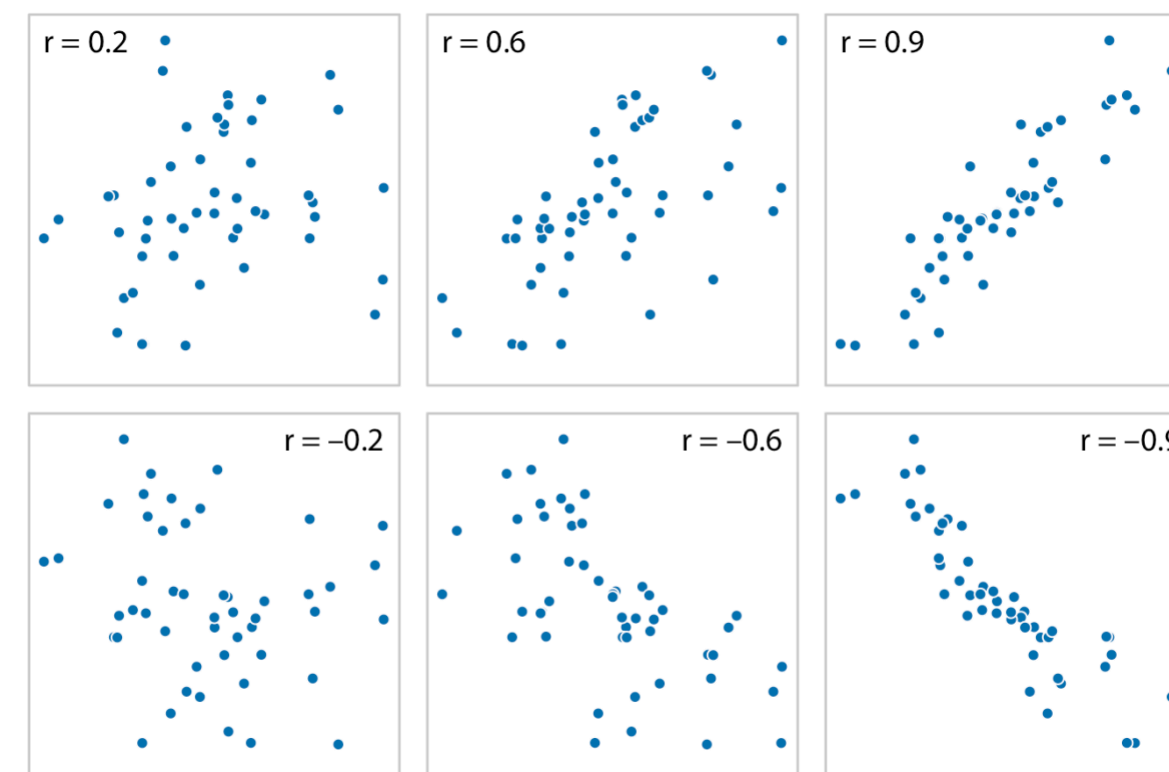
Types of Visualisation: X-Y relationships

- Scatter Plot



Head length (measured from the tip of the bill to the back of the head, in mm) versus body mass (in gram), for 123 blue jays. Each dot corresponds to one bird. There is a moderate tendency for heavier birds to have longer heads. Data source: Keith Tarvin, Oberlin College

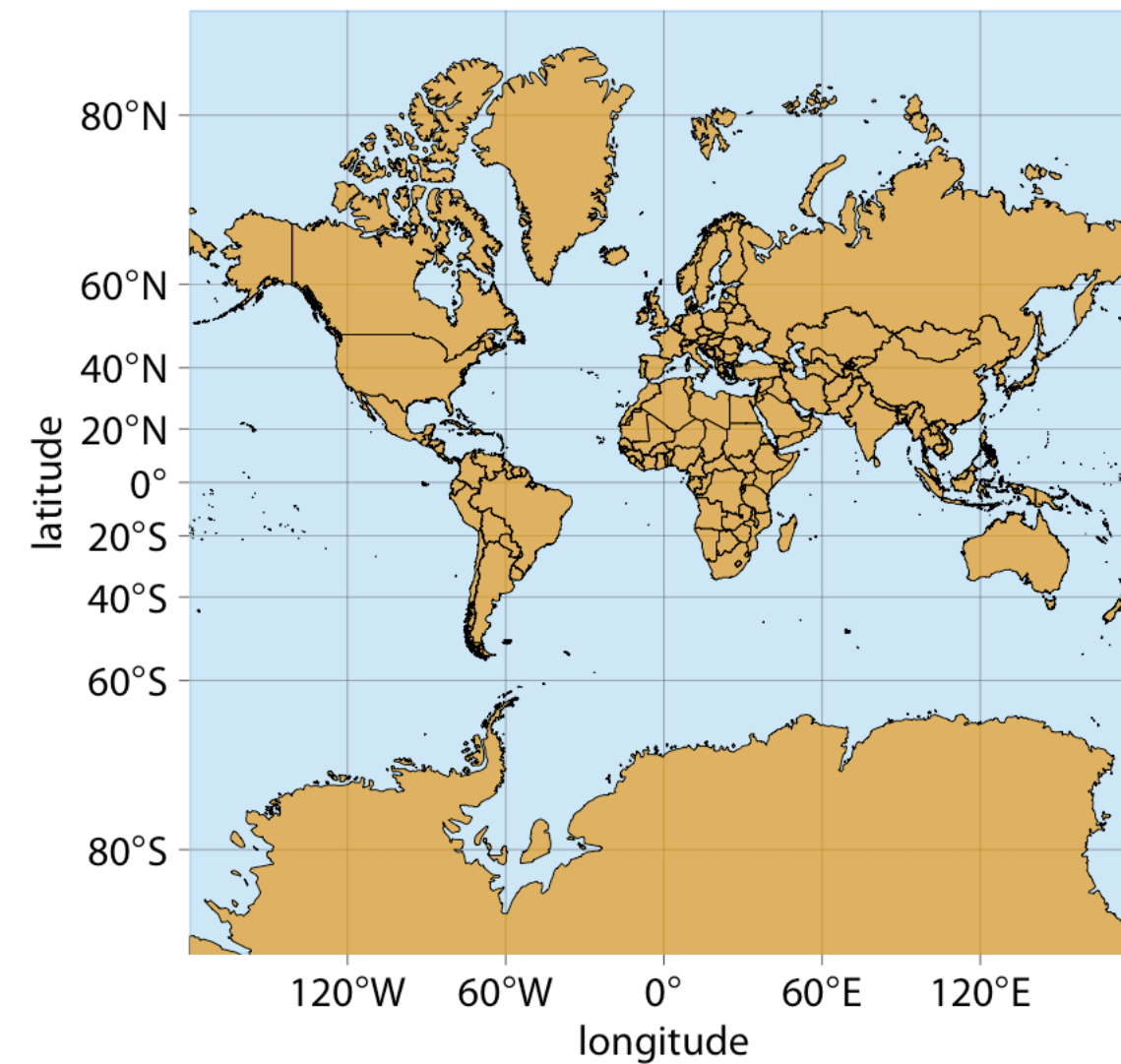
- Correlograms



Examples of correlations of different magnitude and direction, with associated correlation coefficient r . In both rows, from left to right correlations go from weak to strong. In the top row the correlations are positive (larger values for one quantity are associated with larger values for the other) and in the bottom row they are negative (larger values for one quantity are associated with smaller values for the other). In all six panels, the sets of x and y values are identical, but the pairings between individual x and y values have been reshuffled to generate the specified correlation coefficients.

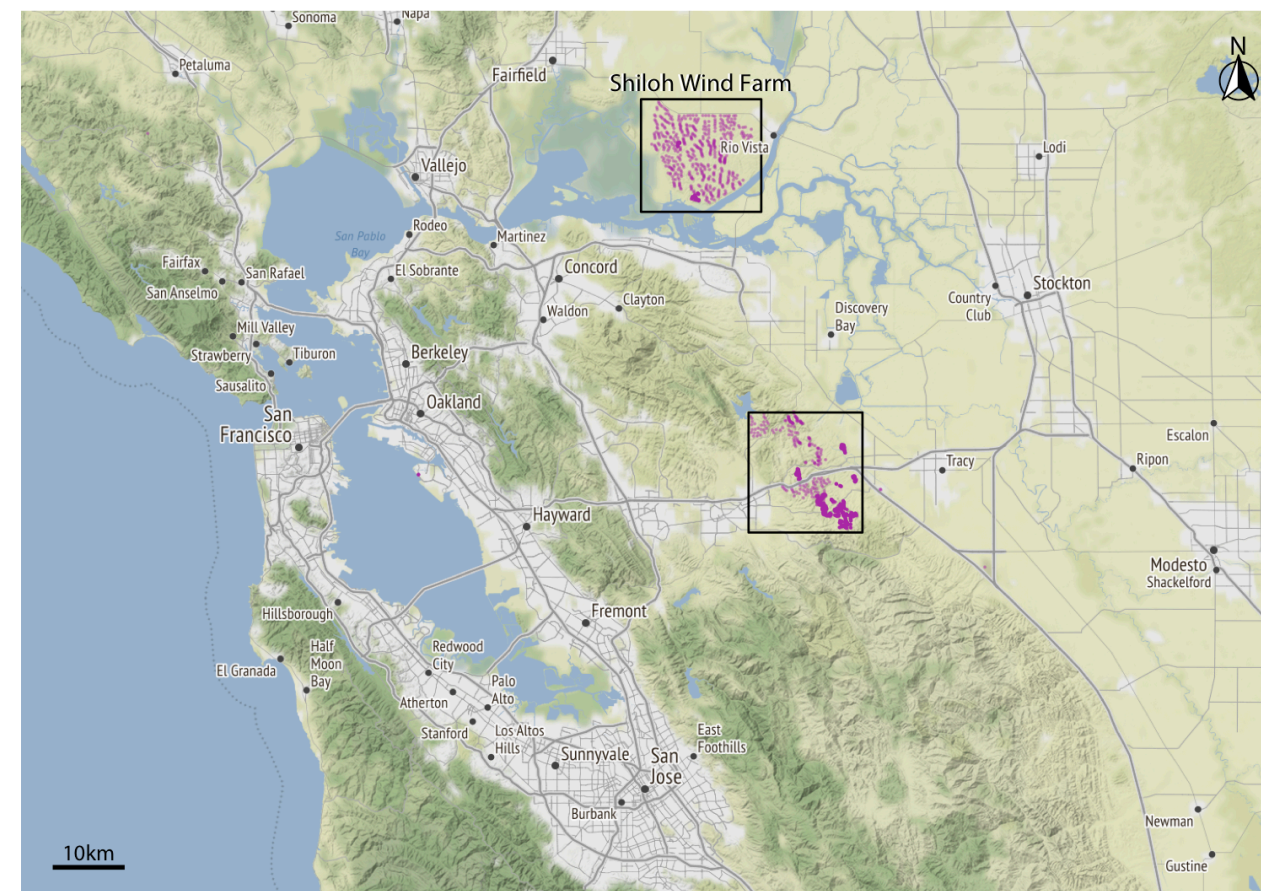
Types of Visualisation: Geospatial Data

- Projections



Mercator projection of the world. In this projection, parallels are straight horizontal lines and meridians are straight vertical lines. It is a conformal projection preserving local angles, but it introduces severe distortions in areas near the poles.

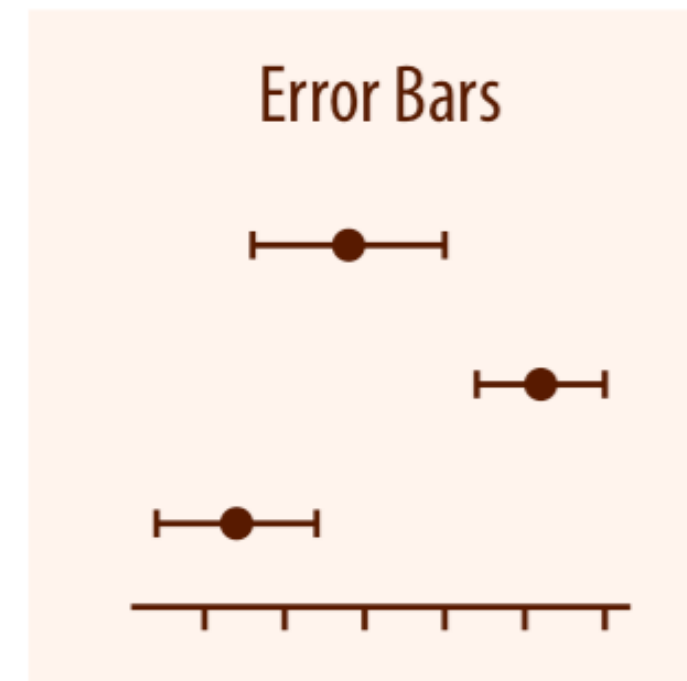
- Layers



Wind turbines in the San Francisco Bay Area. Individual wind turbines are shown as purple-colored dots. Two regions with a high concentration of wind turbines are highlighted with black rectangles. I refer to the wind turbines near Rio Vista collectively as the Shiloh Wind Farm. Map tiles by Stamen Design, under CC BY 3.0. Map data by OpenStreetMap, under ODbL. Wind turbine data: United States Wind Turbine Database

Types of Visualisation: Uncertainty

- Error Bar



- Confidence Strips

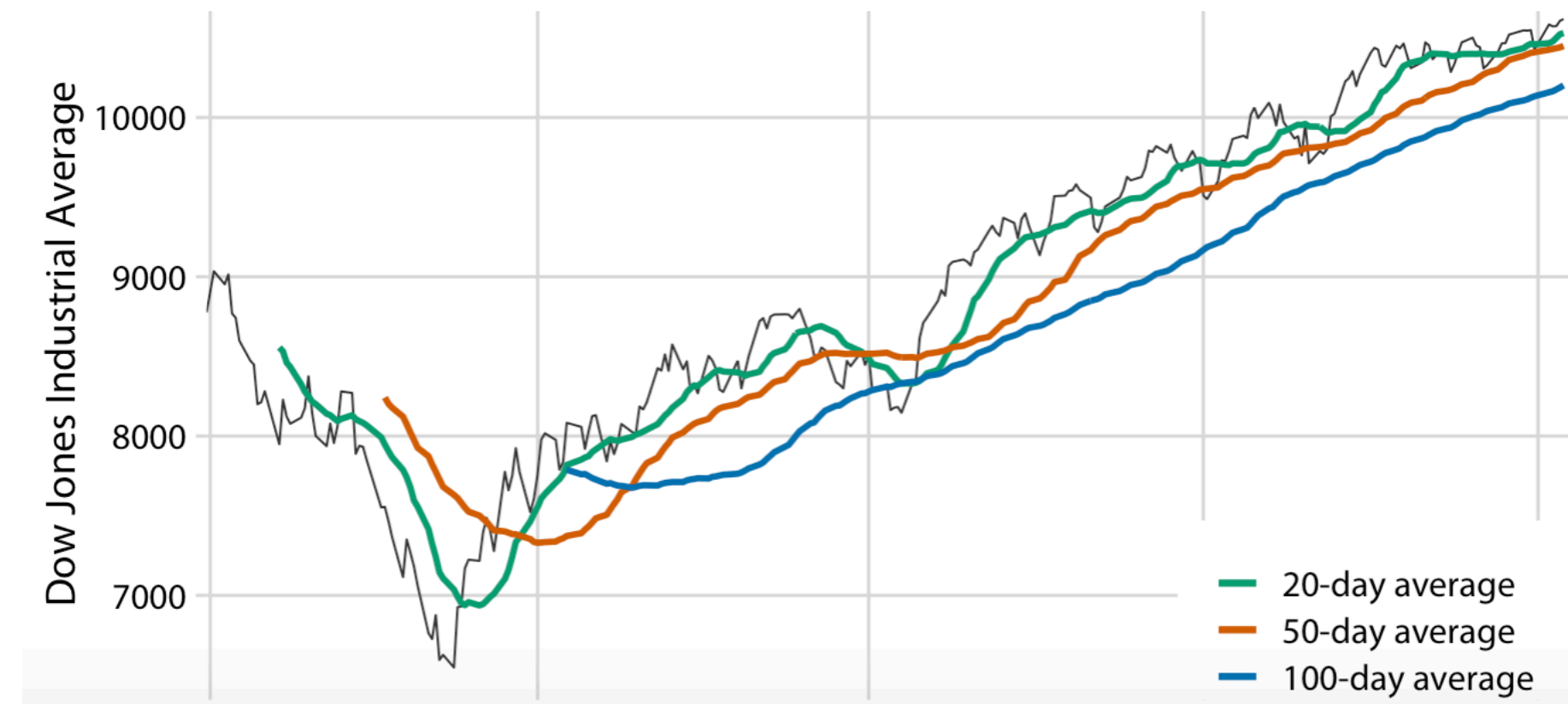


- Confidence Band



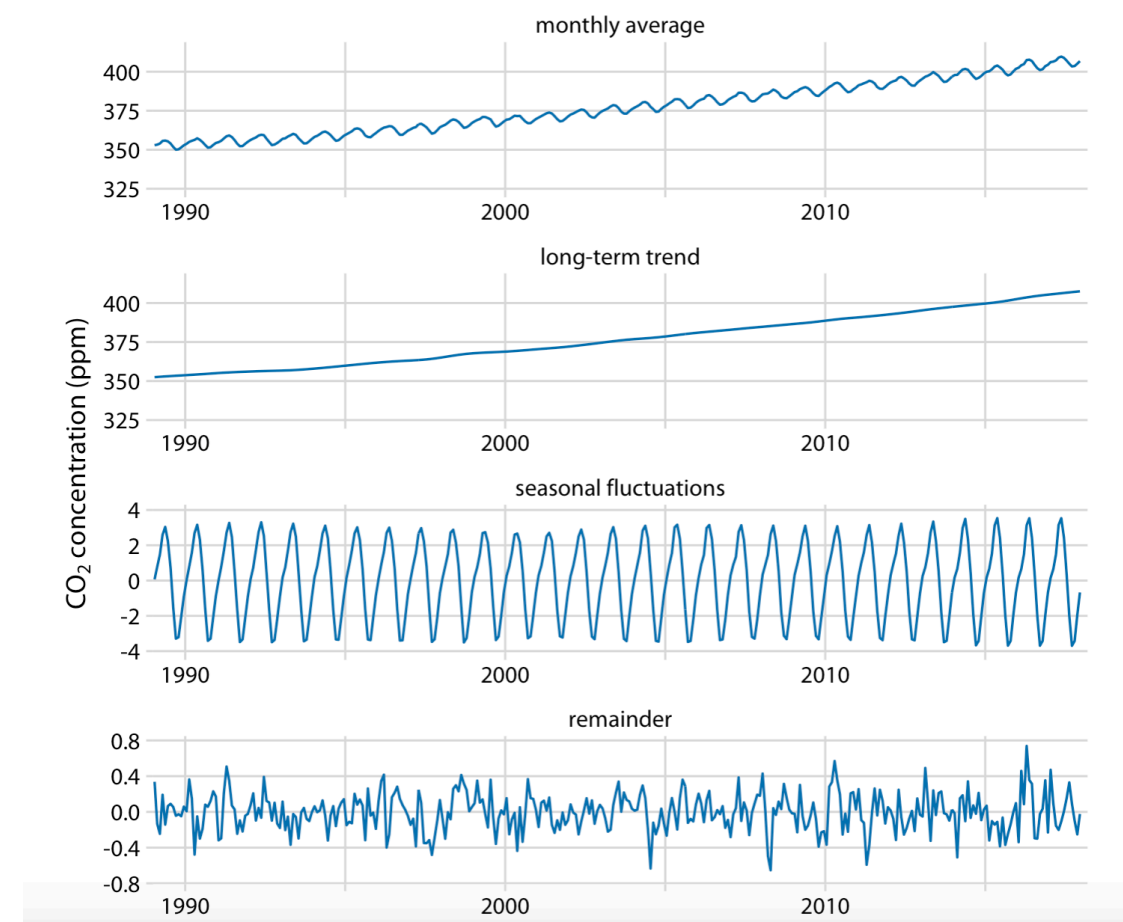
Types of Visualisation: Trends

- Smoothing



Daily closing values of the Dow Jones Industrial Average for the year 2009, shown together with their 20-day, 50-day, and 100-day moving averages. (a) The moving averages are plotted at the end of the moving time windows. (b) The moving averages are plotted in the center of the moving time windows. Data source: Yahoo! Finance

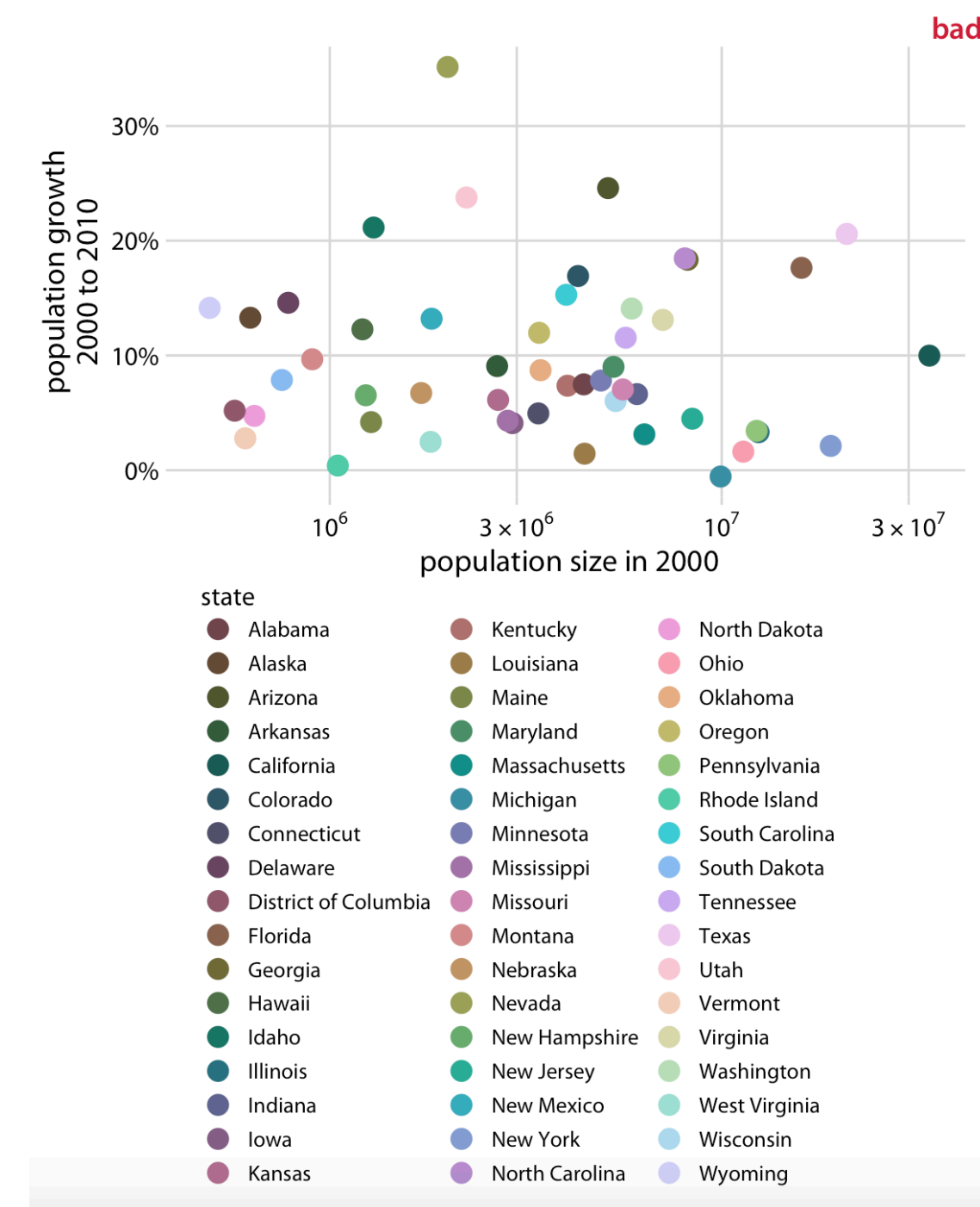
- Time Series



Time-series decomposition of the Keeling curve, showing the monthly average (as in Figure 14.12), the long-term trend, seasonal fluctuations, and the remainder. The remainder is the difference between the actual readings and the sum of the long-term trend and the seasonal fluctuations, and it represents random noise. I have zoomed into the most recent 30 years of data to more clearly show the shape of the annual fluctuations. Data source: Dr. Pieter Tans, NOAA/ESRL, and Dr. Ralph Keeling, Scripps Institution of Oceanography

Common Pitfalls for Color Coding: Part 1

- Encoding too much or irrelevant information
 - As a rule of thumb, qualitative color scales work best when there are three to five different categories that need to be colored. Once we reach eight to ten different categories or more, the task of matching colors to categories becomes too burdensome to be useful, even if the colors remain sufficiently different to be distinguishable in principle.



Population growth from 2000 to 2010 versus population size in 2000, for all 50 U.S. states and the District of Columbia. Every state is marked in a different color. Because there are so many states, it is very difficult to match the colors in the legend to the dots in the scatter plot. Data source: U.S. Census Bureau

it is probably best to use color only to indicate the geographic region of each state and to identify individual states by direct labeling, i.e., by placing appropriate text labels adjacent to the data points

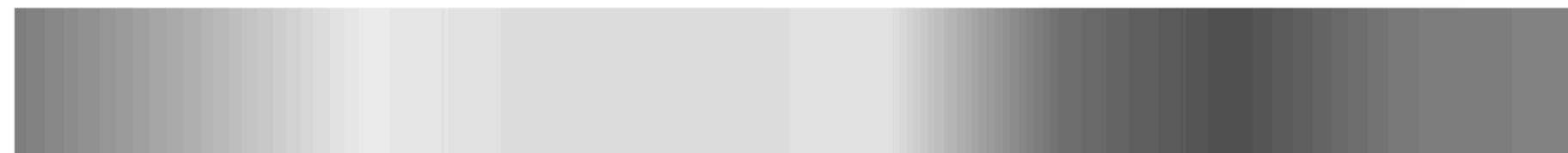
Common Pitfalls for Color Coding: Part 2

- Using non-monotonic color scales to encode data values

rainbow scale



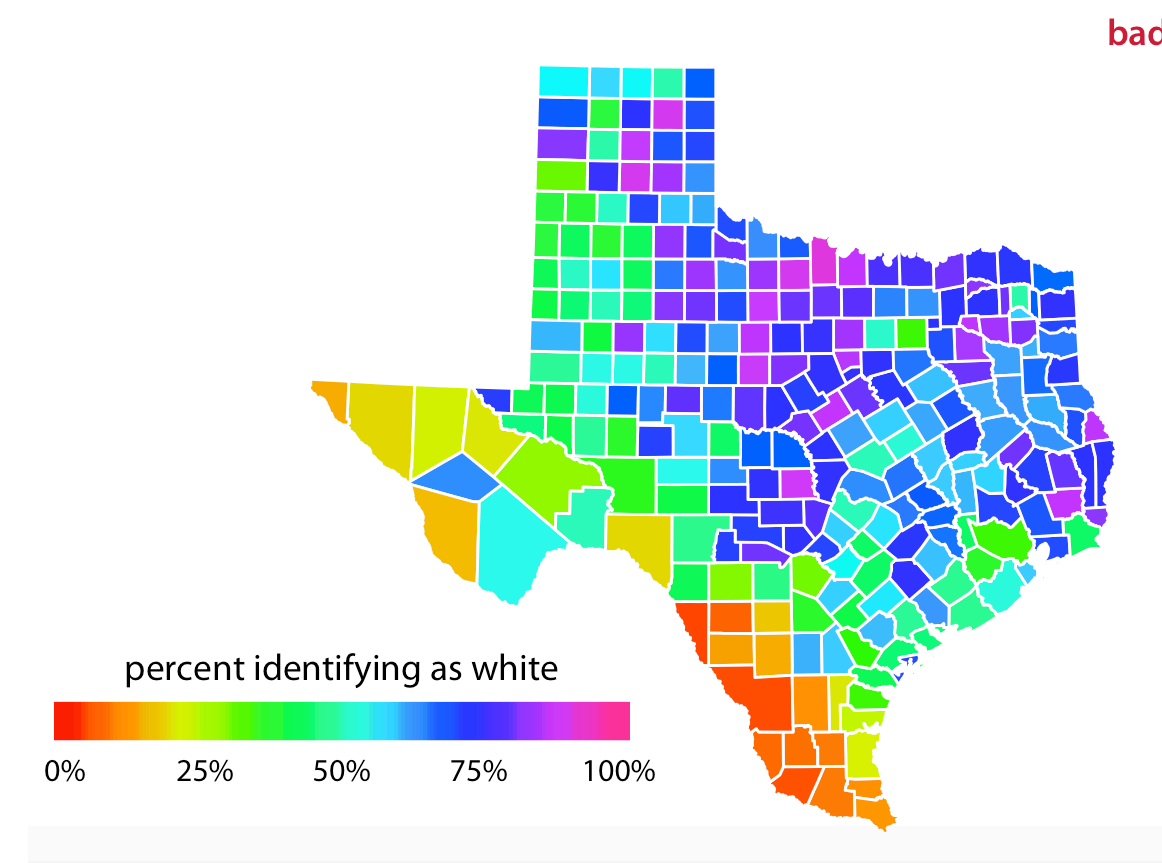
rainbow converted to grayscale



The rainbow colorscale is highly non-monotonic. This becomes clearly visible by converting the colors to gray values. From left to right, the scale goes from moderately dark to light to very dark and back to moderately dark. In addition, the changes in lightness are very non-uniform. The lightest part of the scale (corresponding to the colors yellow, light green, and cyan) takes up almost a third of the entire scale while the darkest part (corresponding to dark blue) is concentrated in a narrow region of the scale.

Common Pitfalls for Color Coding: Part 3

- Not designing for color-vision deficiency



Percentage of people identifying as white in Texas counties. The rainbow color scale is not an appropriate scale to visualize continuous data values, because it tends to place emphasis on arbitrary features of the data. Here, it emphasizes counties in which approximately 75% of the population identify as white. Data source: 2010 Decennial U.S. Census

Python Visualisation Library

- Matplotlib
- Seaborn
- Plotting
- Bokeh
- Pygal
- Geoplotlib